

TAYLOR PENTZ

Senior Software Engineer

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Professional Summary

Senior Software Engineer with 10+ years building **SaaS, healthcare, fintech, and enterprise workflow platforms** using **React 15–18, Node.js 6–18, TypeScript, Express, NestJS, PostgreSQL, MongoDB, AWS, and CI/CD**, delivering scalable front-end systems, resilient APIs, legacy modernization, production debugging, mentoring, architecture ownership, and measurable product value for distributed engineering teams.

Technical Skills

- ❖ **Frontend Engineering:** **React 15–18**, TypeScript, JavaScript ES6+, Redux, Redux Toolkit, React Query, Zustand, Next.js, HTML5, CSS3, SASS, Tailwind CSS, Material UI, responsive UI architecture, accessibility, reusable component systems.
- ❖ **Backend Engineering:** **Node.js 6–18**, Express.js, NestJS, REST APIs, GraphQL, WebSockets, JWT, OAuth2, microservices, event-driven services, background jobs, API versioning, rate limiting, authentication and authorization.
- ❖ **Cloud & DevOps:** **AWS**, ECS, Lambda, S3, CloudFront, API Gateway, RDS, CloudWatch, Docker, Kubernetes basics, GitHub Actions, Jenkins, GitLab CI, Terraform basics, CI/CD pipelines, blue-green deployments.
- ❖ **Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB basics, schema design, indexing, query optimization, migrations, data validation, transactional workflows.
- ❖ **Testing & Quality:** Jest, React Testing Library, Cypress, Playwright, Mocha, Chai, Supertest, unit testing, integration testing, API contract testing, regression testing, code reviews, SonarQube.
- ❖ **Monitoring & Engineering Practices:** Datadog, New Relic, Sentry, CloudWatch Logs, OpenTelemetry basics, Agile/Scrum, architecture documentation, production support, incident triage, mentoring, secure coding, performance tuning.

Professional Experience

SumatoSoft

Senior Software Engineer

Boston, MA (Remote) | Apr 2023 – Mar 2025

- Architected customer-facing **React 18.x** and **Node.js 18.x** product modules for SaaS workflow platforms, improving maintainability through modular UI boundaries, reusable API clients, and clearly documented service contracts.
- Modernized legacy JavaScript screens into **React**, **TypeScript**, and **Redux Toolkit** features, reducing duplicated front-end logic and improving release confidence across high-traffic dashboard workflows.
- Designed and maintained **Express.js** and **NestJS** backend services for account management, reporting, notification, and billing workflows, using **PostgreSQL**, **Redis**, and structured validation layers for reliable request handling.

- Improved page rendering, API response time, and client-side caching behavior through query optimization, lazy loading, memoization, and React Query patterns, increasing key workflow performance by **38%**.
- Resolved production issues involving stale access tokens, duplicate webhook retries, failed background jobs, and inconsistent pagination by adding idempotency keys, retry controls, better logging, and regression coverage.
- Led migration work from loosely organized REST handlers into layered service modules with controller, service, repository, and validation boundaries, making complex business rules easier to test and extend.
- Implemented new customer onboarding, role-based access control, audit history, and export features using **React**, **Node.js**, **PostgreSQL**, and **AWS S3**, supporting smoother enterprise adoption.
- Strengthened CI/CD pipelines with **GitHub Actions**, Docker image validation, automated test gates, and environment-specific deployment checks, reducing missed build issues before staging releases.
- Partnered with product managers, QA engineers, and designers to break large epics into deliverable increments, review edge cases early, and mentor mid-level engineers on scalable React and API patterns.
- Improved operational visibility with **Sentry**, **CloudWatch**, structured logs, and dashboard alerts, enabling faster investigation of authentication errors, slow database queries, and failed integration calls.

Hexaview Technologies Inc.

Software Engineer

New York, NY | Jul 2018 – Mar 2023

- Developed enterprise web applications using **React 16–17**, **Node.js 12–16**, **Express.js**, **TypeScript**, and **MongoDB**, supporting financial, healthcare, and internal operations platforms with complex business workflows.
- Migrated older class-based React components into functional components with hooks, cleaner state handling, and shared UI libraries, improving development speed and reducing front-end defects by **31%**.
- Built REST and GraphQL API layers for reporting, document management, user preferences, payment workflows, and notification services, applying authentication, validation, pagination, and error handling consistently.
- Resolved production defects involving memory leaks in long-running Node.js workers, slow MongoDB aggregation queries, race conditions in status updates, and inconsistent file-upload retry behavior.
- Modernized several Node.js services from callback-heavy patterns to async/await, centralized middleware, structured logging, and reusable database access utilities for easier maintenance.
- Designed and deployed scalable backend and frontend services on **Microsoft Azure**, including Azure App Service, Azure Functions, and Azure Storage, ensuring high availability, secure configuration, and performance optimization for enterprise applications.
- Implemented dashboard filtering, bulk-edit workflows, user activity timelines, and admin permission features using **React**, **Redux**, **Material UI**, and backend authorization rules.
- Implemented CI/CD pipelines using **Azure DevOps**, integrating automated builds, testing, and release management, while leveraging Azure Monitor and Application Insights for centralized logging, alerting, and production observability.
- Collaborated with QA teams to expand **Jest**, **Cypress**, and API integration coverage around checkout flows, user management, reporting exports, and regression-prone legacy screens.
- Supported architecture discussions, code reviews, sprint planning, and production triage while mentoring junior engineers on React component design, REST API quality, and debugging practices.

SimbirSoft Company

Software Developer

Boston, MA | Mar 2015 – Jun 2018

- Built web application features across early **Node.js 6.x**, **Express.js**, jQuery, Backbone, and later **React 15.x** modules, supporting customer portals, internal dashboards, and operations tools.
- Introduced newer React components into legacy front-end pages where full rewrites were not practical, improving user interaction quality and reducing repeated UI bugs by **24%**.
- Created backend endpoints for user profiles, search screens, file metadata, approval workflows, and reporting tools using **Node.js**, **Express**, **MySQL**, and **MongoDB**.
- Fixed recurring production issues involving broken form validation, inconsistent date formatting, session timeout errors, duplicate records, and slow search queries caused by missing database indexes.
- Helped migrate selected JavaScript modules from scattered script files into organized application folders with shared utilities, clearer naming, and reusable API communication patterns.
- Implemented new features for admin dashboards, CSV exports, customer lookup, and activity logging while balancing older jQuery code with newer React-based interface sections.
- Improved application reliability by adding unit tests, manual regression checklists, clearer deployment notes, and better error messages for support teams troubleshooting user-reported issues.
- Worked closely with senior engineers during database schema updates, release planning, bug verification, and production hotfixes, building strong foundations in full-stack delivery.

Microsoft

Software Developer Intern

Cambridge, MA | Aug 2014 – Feb 2015

- Supported internal engineering tools using **JavaScript**, early **Node.js 0.10**, C#, ASP.NET MVC, and SQL Server, helping teams automate repetitive reporting and data-review workflows.
- Built small backend utilities for internal APIs, scheduled jobs, and data cleanup tasks, learning how production systems required validation, logging, permission checks, and predictable failure handling.
- Assisted with front-end fixes in internal dashboards by improving form behavior, table filtering, browser compatibility, and client-side validation for operations users.
- Investigated bugs involving inconsistent data refreshes, malformed API responses, null-value exceptions, and incorrect status labels by tracing logs and reproducing issues in test environments.
- Created documentation for setup steps, known limitations, release notes, and troubleshooting guidance so future interns and engineers could support the tools more easily.
- Participated in code reviews, sprint ceremonies, and technical discussions with senior engineers, gaining practical exposure to enterprise software quality standards.
- Delivered internship features with careful testing, peer feedback, and incremental commits, building a strong base in maintainable full-stack engineering practices.

Projects

• Enterprise Workflow Modernization Platform

Designed a modernization project that moved legacy dashboard workflows into **React 18**, **TypeScript**, **Node.js**, **Express**, and **PostgreSQL**, adding reusable UI components, structured API contracts, role-based access, audit logging, and deployment checks that made future feature development faster and safer.

- **Real-Time Operations Reporting System**

Built a reporting and notification platform using **React**, **Node.js**, **Redis**, **WebSockets**, **AWS S3**, and **PostgreSQL**, supporting live status updates, export processing, background jobs, searchable activity history, and production monitoring for business operations teams.

Certifications

- **AWS Certified Developer – Associate**
- **AWS Certified Solutions Architect – Associate**
- **MongoDB Developer Certification**
- **Meta Front-End Developer Professional Certificate**
- **Scrum Alliance Certified ScrumMaster**

Education

University of North Carolina at Chapel Hill

Bachelor's Degree in Computer Science (2010 – 2014)